

* Rank 0 tensor. * Rank 1 tensor. * Rank 2 tensor (s).

$$\begin{bmatrix} 0 & M_{xy} & 0 \\ M_{xy} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{xy} = \mathbb{M}_{xy}^{(1)}$$

* Rank 2 tensor (a).

$$\begin{bmatrix} 0 & M_{xy} & 0 \\ -M_{xy} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{xy} = \mathbb{T}_z^{(1)}$$

* Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ M_{yzx} & 0 & 0 \\ 0 & M_{yzx} & 0 \\ 0 & 0 & M_{yzx} \end{bmatrix}$$

$$M_{yzx} = \mathbb{M}_{xyz}^{(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ M_{yzx} & 0 & 0 \\ 0 & M_{zxy} & 0 \\ 0 & 0 & -M_{yzx} - M_{zxy} \end{bmatrix}$$

$$M_{yzx} = 3\mathbb{T}_u^{(1)} + \mathbb{T}_v^{(1)}$$

$$M_{zxy} = -3\mathbb{T}_u^{(1)} + \mathbb{T}_v^{(1)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} M_{yzx} & 0 & 0 \\ 0 & M_{zxy} & 0 \\ 0 & 0 & M_{xyz} \end{bmatrix}$$

$$M_{yzx} = \mathbb{T}_0^{(1)} - \mathbb{T}_u^{(2)} + \mathbb{T}_v^{(2)}$$

$$M_{zxy} = \mathbb{T}_0^{(1)} - \mathbb{T}_u^{(2)} - \mathbb{T}_v^{(2)}$$

$$M_{xyz} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_u^{(2)}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & M_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & M_{yyxy} \\ 0 & 0 & 0 & 0 & 0 & M_{zzxy} \\ 0 & 0 & 0 & 0 & M_{yzzx} & 0 \\ 0 & 0 & 0 & M_{yzzx} & 0 & 0 \\ M_{xxxy} & M_{yyxy} & M_{zzxy} & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned}
M_{xxxy} &= \mathbb{M}_{4z}^{\alpha(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} - \mathbb{M}_z^{\beta(1)} \\
M_{yyxy} &= -\mathbb{M}_{4z}^{\alpha(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} - \mathbb{M}_z^{\beta(1)} \\
M_{zzxy} &= \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_z^{\beta(1)} \\
M_{yzzx} &= \mathbb{M}_{xy}^{(2)} + 2\mathbb{M}_z^{\beta(1)}
\end{aligned}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & M_{xxxy} \\
0 & 0 & 0 & 0 & 0 & M_{yyxy} \\
0 & 0 & 0 & 0 & 0 & M_{zzxy} \\
0 & 0 & 0 & 0 & M_{yzzx} & 0 \\
0 & 0 & 0 & -M_{yzzx} & 0 & 0 \\
-M_{xxxy} & -M_{yyxy} & -M_{zzxy} & 0 & 0 & 0
\end{bmatrix}$$

$$\begin{aligned}
M_{xxxy} &= \mathbb{M}_{xy}^{(3)} + 2\mathbb{T}_z^{(1)} - 2\mathbb{T}_z^{\alpha(1)} + 2\mathbb{T}_z^{\beta(1)} \\
M_{yyxy} &= \mathbb{M}_{xy}^{(3)} - 2\mathbb{T}_z^{(1)} + 2\mathbb{T}_z^{\alpha(1)} + 2\mathbb{T}_z^{\beta(1)} \\
M_{zzxy} &= \mathbb{M}_{xy}^{(3)} - 4\mathbb{T}_z^{\beta(1)} \\
M_{yzzx} &= -\mathbb{T}_z^{(1)} - 4\mathbb{T}_z^{\alpha(1)}
\end{aligned}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix}
0 & M_{yzzx} & 0 \\
M_{yzzx} & 0 & 0 \\
0 & 0 & 0
\end{bmatrix}$$

$$M_{yzzx} = 2\mathbb{M}_{xy}^{(4)}$$

* Rank 4 tensor (aaa).

$$\begin{bmatrix}
0 & M_{yzzx} & 0 \\
-M_{yzzx} & 0 & 0 \\
0 & 0 & 0
\end{bmatrix}$$

$$M_{yzzx} = -\mathbb{T}_z^{(2)}$$

* Rank 4 tensor (sa).

$$\begin{bmatrix}
0 & 0 & M_{xxxy} \\
0 & 0 & M_{yyxy} \\
0 & 0 & M_{zzxy} \\
0 & M_{yzzx} & 0 \\
M_{zxyz} & 0 & 0 \\
0 & 0 & 0
\end{bmatrix}$$

$$\begin{aligned}
M_{xxxy} &= -2\mathbb{M}_{xy}^{(5)} + \mathbb{T}_z^{(3)} - \mathbb{T}_z^{\alpha(2)} + \mathbb{T}_z^{\beta(2)} \\
M_{yyxy} &= 2\mathbb{M}_{xy}^{(5)} + \mathbb{T}_z^{(3)} - \mathbb{T}_z^{\alpha(2)} - \mathbb{T}_z^{\beta(2)} \\
M_{zzxy} &= \mathbb{T}_z^{(3)} + 2\mathbb{T}_z^{(4)} + 2\mathbb{T}_z^{\alpha(2)} \\
M_{yzzx} &= -\mathbb{M}_{xy}^{(5)} + \mathbb{T}_z^{(4)} - \mathbb{T}_z^{\alpha(2)} - \mathbb{T}_z^{\beta(2)} \\
M_{zxyz} &= \mathbb{M}_{xy}^{(5)} + \mathbb{T}_z^{(4)} - \mathbb{T}_z^{\alpha(2)} + \mathbb{T}_z^{\beta(2)}
\end{aligned}$$

* Rank 4 tensor (as).

$$\begin{bmatrix}
0 & 0 & 0 & 0 & M_{yzzx} & 0 \\
0 & 0 & 0 & M_{zxyz} & 0 & 0 \\
M_{xyxx} & M_{xyyy} & M_{xyzx} & 0 & 0 & 0
\end{bmatrix}$$

$$\begin{aligned}
M_{yzzx} &= \mathbb{M}_{xy}^{(6)} + \mathbb{T}_z^{(6)} - \mathbb{T}_z^{\alpha(3)} + \mathbb{T}_z^{\beta(3)} \\
M_{zxyz} &= -\mathbb{M}_{xy}^{(6)} + \mathbb{T}_z^{(6)} - \mathbb{T}_z^{\alpha(3)} - \mathbb{T}_z^{\beta(3)} \\
M_{xyxx} &= -2\mathbb{M}_{xy}^{(6)} + \mathbb{T}_z^{(5)} - \mathbb{T}_z^{\alpha(3)} + \mathbb{T}_z^{\beta(3)} \\
M_{xyyy} &= 2\mathbb{M}_{xy}^{(6)} + \mathbb{T}_z^{(5)} - \mathbb{T}_z^{\alpha(3)} - \mathbb{T}_z^{\beta(3)} \\
M_{xyzx} &= \mathbb{T}_z^{(5)} + 2\mathbb{T}_z^{(6)} + 2\mathbb{T}_z^{\alpha(3)}
\end{aligned}$$

* Rank 4 tensor (s).

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & M_{xxxy} & 0 & 0 & M_{xxyx} \\
0 & 0 & 0 & 0 & 0 & M_{yyxy} & 0 & 0 & M_{yyyx} \\
0 & 0 & 0 & 0 & 0 & M_{zzxy} & 0 & 0 & M_{zzyx} \\
0 & 0 & 0 & 0 & M_{yzxz} & 0 & 0 & M_{yzxz} & 0 \\
0 & 0 & 0 & M_{zxyz} & 0 & 0 & M_{zxzy} & 0 & 0 \\
M_{xyxx} & M_{xyyy} & M_{xyzx} & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$$\begin{aligned}
M_{xxxy} &= \mathbb{M}_{4z}^{\alpha(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(3)} - 2\mathbb{M}_{xy}^{(5)} - \mathbb{M}_z^{\beta(1)} + 2\mathbb{T}_z^{(1)} + \mathbb{T}_z^{(3)} - 2\mathbb{T}_z^{\alpha(1)} - \mathbb{T}_z^{\alpha(2)} + 2\mathbb{T}_z^{\beta(1)} + \mathbb{T}_z^{\beta(2)} \\
M_{xxyx} &= \mathbb{M}_{4z}^{\alpha(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(3)} + 2\mathbb{M}_{xy}^{(5)} - \mathbb{M}_z^{\beta(1)} + 2\mathbb{T}_z^{(1)} - \mathbb{T}_z^{(3)} - 2\mathbb{T}_z^{\alpha(1)} + \mathbb{T}_z^{\alpha(2)} + 2\mathbb{T}_z^{\beta(1)} - \mathbb{T}_z^{\beta(2)} \\
M_{yyxy} &= -\mathbb{M}_{4z}^{\alpha(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(3)} + 2\mathbb{M}_{xy}^{(5)} - \mathbb{M}_z^{\beta(1)} - 2\mathbb{T}_z^{(1)} + \mathbb{T}_z^{(3)} + 2\mathbb{T}_z^{\alpha(1)} - \mathbb{T}_z^{\alpha(2)} + 2\mathbb{T}_z^{\beta(1)} - \mathbb{T}_z^{\beta(2)} \\
M_{yyyx} &= -\mathbb{M}_{4z}^{\alpha(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(3)} - 2\mathbb{M}_{xy}^{(5)} - \mathbb{M}_z^{\beta(1)} - 2\mathbb{T}_z^{(1)} - \mathbb{T}_z^{(3)} + 2\mathbb{T}_z^{\alpha(1)} + \mathbb{T}_z^{\alpha(2)} + 2\mathbb{T}_z^{\beta(1)} + \mathbb{T}_z^{\beta(2)} \\
M_{zzxy} &= \mathbb{M}_{xy}^{(1)} + \mathbb{M}_{xy}^{(3)} + 2\mathbb{M}_z^{\beta(1)} + \mathbb{T}_z^{(3)} + 2\mathbb{T}_z^{(4)} + 2\mathbb{T}_z^{\alpha(2)} - 4\mathbb{T}_z^{\beta(1)} \\
M_{zzyx} &= \mathbb{M}_{xy}^{(1)} + \mathbb{M}_{xy}^{(3)} + 2\mathbb{M}_z^{\beta(1)} - \mathbb{T}_z^{(3)} - 2\mathbb{T}_z^{(4)} - 2\mathbb{T}_z^{\alpha(2)} - 4\mathbb{T}_z^{\beta(1)} \\
M_{yzxz} &= \mathbb{M}_{xy}^{(2)} - \mathbb{M}_{xy}^{(5)} + 2\mathbb{M}_z^{\beta(1)} - \mathbb{T}_z^{(1)} + \mathbb{T}_z^{(4)} - 4\mathbb{T}_z^{\alpha(1)} - \mathbb{T}_z^{\alpha(2)} - \mathbb{T}_z^{\beta(2)} \\
M_{yzxz} &= \mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(5)} + 2\mathbb{M}_z^{\beta(1)} - \mathbb{T}_z^{(1)} - \mathbb{T}_z^{(4)} - 4\mathbb{T}_z^{\alpha(1)} + \mathbb{T}_z^{\alpha(2)} + \mathbb{T}_z^{\beta(2)} \\
M_{zxyz} &= \mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(5)} + 2\mathbb{M}_z^{\beta(1)} + \mathbb{T}_z^{(1)} + \mathbb{T}_z^{(4)} + 4\mathbb{T}_z^{\alpha(1)} - \mathbb{T}_z^{\alpha(2)} + \mathbb{T}_z^{\beta(2)} \\
M_{zxzy} &= \mathbb{M}_{xy}^{(2)} - \mathbb{M}_{xy}^{(5)} + 2\mathbb{M}_z^{\beta(1)} + \mathbb{T}_z^{(1)} - \mathbb{T}_z^{(4)} + 4\mathbb{T}_z^{\alpha(1)} + \mathbb{T}_z^{\alpha(2)} - \mathbb{T}_z^{\beta(2)} \\
M_{xyxx} &= \mathbb{M}_{4z}^{\alpha(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} - \mathbb{M}_{xy}^{(3)} - \mathbb{M}_z^{\beta(1)} - 2\mathbb{T}_z^{(1)} + 2\mathbb{T}_z^{\alpha(1)} - 2\mathbb{T}_z^{\beta(1)} \\
M_{xyyy} &= -\mathbb{M}_{4z}^{\alpha(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} - \mathbb{M}_{xy}^{(3)} - \mathbb{M}_z^{\beta(1)} + 2\mathbb{T}_z^{(1)} - 2\mathbb{T}_z^{\alpha(1)} - 2\mathbb{T}_z^{\beta(1)} \\
M_{xyzx} &= \mathbb{M}_{xy}^{(1)} - \mathbb{M}_{xy}^{(3)} + 2\mathbb{M}_z^{\beta(1)} + 4\mathbb{T}_z^{\beta(1)}
\end{aligned}$$

* Rank 4 tensor (a).

$$\begin{bmatrix}
0 & 0 & 0 & 0 & M_{yzxz} & 0 & 0 & M_{yzxz} & 0 \\
0 & 0 & 0 & M_{zxyz} & 0 & 0 & M_{zxzy} & 0 & 0 \\
M_{xyxx} & M_{xyyy} & M_{xyzx} & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$$\begin{aligned}
M_{yzxz} &= 2\mathbb{M}_{xy}^{(4)} + \mathbb{M}_{xy}^{(6)} - \mathbb{T}_z^{(2)} + \mathbb{T}_z^{(6)} - \mathbb{T}_z^{\alpha(3)} + \mathbb{T}_z^{\beta(3)} \\
M_{yzxz} &= -2\mathbb{M}_{xy}^{(4)} + \mathbb{M}_{xy}^{(6)} + \mathbb{T}_z^{(2)} + \mathbb{T}_z^{(6)} - \mathbb{T}_z^{\alpha(3)} + \mathbb{T}_z^{\beta(3)} \\
M_{zxyz} &= 2\mathbb{M}_{xy}^{(4)} - \mathbb{M}_{xy}^{(6)} + \mathbb{T}_z^{(2)} + \mathbb{T}_z^{(6)} - \mathbb{T}_z^{\alpha(3)} - \mathbb{T}_z^{\beta(3)} \\
M_{zxzy} &= -2\mathbb{M}_{xy}^{(4)} - \mathbb{M}_{xy}^{(6)} - \mathbb{T}_z^{(2)} + \mathbb{T}_z^{(6)} - \mathbb{T}_z^{\alpha(3)} - \mathbb{T}_z^{\beta(3)} \\
M_{xyxx} &= -2\mathbb{M}_{xy}^{(6)} + \mathbb{T}_z^{(5)} - \mathbb{T}_z^{\alpha(3)} + \mathbb{T}_z^{\beta(3)} \\
M_{xyyy} &= 2\mathbb{M}_{xy}^{(6)} + \mathbb{T}_z^{(5)} - \mathbb{T}_z^{\alpha(3)} - \mathbb{T}_z^{\beta(3)} \\
M_{xyzx} &= \mathbb{T}_z^{(5)} + 2\mathbb{T}_z^{(6)} + 2\mathbb{T}_z^{\alpha(3)}
\end{aligned}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} 0 & M_{xxxy} & 0 \\ M_{yyyyx} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & M_{zzxy} & 0 \\ M_{xxyx} & 0 & 0 \\ M_{yzzx} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & M_{xyyy} & 0 \\ 0 & 0 & \frac{M_{xxxx}}{2} - \frac{M_{xxyx}}{2} + M_{xyyy} - M_{yyyx} + M_{zzxy} \end{bmatrix}$$

$$M_{xxxy} = \mathbb{M}_{4z}^{\alpha(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(3)} - 2\mathbb{M}_{xy}^{(5)} - \mathbb{M}_z^{\beta(1)}$$

$$M_{yyyyx} = -\mathbb{M}_{4z}^{\alpha(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(3)} - 2\mathbb{M}_{xy}^{(5)} - \mathbb{M}_z^{\beta(1)}$$

$$M_{zzxy} = \mathbb{M}_{xy}^{(1)} + \mathbb{M}_{xy}^{(3)} + 2\mathbb{M}_z^{\beta(1)}$$

$$M_{xxyx} = \mathbb{M}_{4z}^{\alpha(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(3)} + 2\mathbb{M}_{xy}^{(5)} - \mathbb{M}_z^{\beta(1)}$$

$$M_{yzzx} = \mathbb{M}_{xy}^{(2)} - \mathbb{M}_{xy}^{(5)} + 2\mathbb{M}_z^{\beta(1)}$$

$$M_{xyyy} = -\mathbb{M}_{4z}^{\alpha(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} - \mathbb{M}_{xy}^{(3)} - \mathbb{M}_z^{\beta(1)}$$